

Margaux Delizée

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Education

- M.Eng. – Biomedical Engineering | Boston University, USA** **Sept. 2026–May 2027**
- BAEF Fellow (Belgian American Educational Foundation) & BU Dean's Graduate Scholarship recipient
- M.S. – Electrical Engineering | University of Mons, Belgium** **Sept. 2024–Jun. 2026**
- Average: 18/20 (*Summa cum laude*).
 - Specialization in Life Sciences Applications.
 - Relevant coursework: Robotics and Biomedical Applications, Medical Image and Signal Processing.
- Exchange program – Electrical Engineering | Polytechnique Montréal, Canada** **Aug. 2024–Dec. 2024**
- 3.9 GPA
 - Relevant coursework: Signal Analysis, Multivariable Systems, Biomedical Measurements.
- B.S. – Electrical Engineering | University of Mons, Belgium** **Sept. 2021–Jun. 2024**
- Graduated with honors, top 10%.
 - Relevant coursework: Physics, Thermodynamics, Electromagnetism, Electronics, Differential Equations, Probability and Statistics, Calculus, Physical and Organic Chemistry.

Research & Academic Projects

- Master's thesis | Lehigh University PA** **Feb. 2026–Jun. 2026**
- Developed closed-loop control algorithms for cardiovascular dynamics regulation via vagal nerve stimulation, targeting atrial fibrillation in a human in silico model.
 - Submitted abstract for poster presentation at SfN Annual Meeting, Nov. 2026 (under review).
 - Manuscript in preparation for journal submission.
- Control Strategies for Drug Delivery | University of Mons** **Feb. 2025–Jun. 2025**
- Designed advanced drug delivery controllers in a surgical patient simulator (MATLAB).
 - Independently authored a research article exploring model-based control approaches.
- ECG Acquisition & Analysis | Polytechnique Montréal** **Sept. 2024–Dec. 2024**
- Recorded and analyzed ECG signals using Biopac.
 - Applied filtering and signal processing techniques to detect cardiac abnormalities.
- Smart Activity Monitoring Device | University of Mons** **Feb. 2024–Jun. 2024**
- Built a wearable fitness tracker with heart rate, accelerometer, and GPS sensors (with Arduino).
- Automated Object Sorting System | University of Mons** **Sept. 2021–Dec. 2021**
- Designed and implemented an automated sorting system integrating hardware and software design.

Professional Experience

- System Engineer Intern | IBA, Louvain-la-Neuve, Belgium** **Jul. 2025–Oct. 2025**
- Contributed to the development of control algorithms for the proton therapy gantry system, enhancing the precision of treatment with the proton beam.
 - Collaborated with engineers to integrate control, software, and biomedical requirements.
 - Performed 3-day on-site testing of control algorithms at a proton therapy center in Reading, UK.
- Mathematics Tutor** **Jan. 2022 – Present**
- Provide mathematics tutoring for high school and university students, including preparation for engineering entrance exams, with a focus on algebra, calculus, and geometry.

Skills

Programming: Python (NumPy, PyTorch), C++, Arduino, MATLAB (Simulink) • Biomedical signal processing (filtering, feature extraction) • Scientific writing & publication • Computational modelling • Interdisciplinary teamwork • Project leadership • French (native), English (C1, TOEFL 103/120).